

Oswald Gyabaah

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PROFESSIONAL SUMMARY

DevOps and Infrastructure Engineering Lead with 5+ years building and owning production systems at scale. Experienced across AWS, GCP, Azure, and on-premise Kubernetes environments (EKS, GKE, TKG, RKE2), with expertise in API design, database modeling and query optimization, CI/CD pipeline automation, and infrastructure tooling development. Led teams delivering high-throughput services handling 3M+ daily requests at 99.97% uptime. Built observability stacks (Prometheus, Grafana, Elastic Stack) and automated infrastructure provisioning with Terraform and Ansible. Currently building infrastructure automation tooling and AI-powered agents for production operations.

CORE COMPETENCIES

Kubernetes at Scale (EKS, GKE, TKG, RKE2)

CI/CD Pipeline Automation

API Design & Microservices

Database Optimization & Schema Design

Infrastructure Tooling & Automation

Production Systems Ownership

Observability & Monitoring

Cloud Infrastructure (AWS, GCP, Azure)

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD

May 2025 - Present

Manager, Finacle Services Team

- Manage **800+ production pods** on Tanzu Kubernetes Grid (TKG) sustaining **1.5M+ daily banking transactions**, overseeing full cluster lifecycle: deployments, patching, and incident response
- Manage the **observability stack** with dashboards tracking CPU/memory trends, transaction volumes, response times, and categorized exceptions -- enabling threshold-based alerting and proactive scaling
- Lead L2 support operations, correlating transactional and application logs across interfaces and **core banking microservices** via the Elastic stack, driving root-cause analysis and cross-functional remediation
- Led end-to-end QA for core banking migration: planned and executed **10,000+ test cases** across 13 modules and 33 applications

Guaranty Trust Bank Ghana LTD

Apr 2024 - Present

DevOps Engineering Lead

- Architected **high-availability Kubernetes clusters** across multiple on-prem environments handling **3M+ daily requests with 99.97% uptime SLA**; implemented auto-scaling and self-healing mechanisms
- Implemented **CI/CD automation** that cut deployment time by ~90% and increased release velocity ~5x; integrated Veracode and Trivy for automated security analysis
- Introduced unified **observability stack** (Prometheus, Grafana, Elastic Stack, Sentry) for centralized logging, tracing, and real-time alerting to accelerate RCA
- Implemented stateless, version-controlled cluster deployments with FluxCD, Helm, and Azure DevOps for consistency and rapid **disaster recovery** rebuilds
- Founded the bank's DevOps team, leading engineers to establish modern delivery practices -- Kubernetes, containerized deployments, and automated release workflows
- Manage **multi-cloud infrastructure** across Azure and AWS: resource provisioning, private VPC links to on-prem systems, access control, and cost optimization

Chief Technical Officer

- Architected the full platform end-to-end: requirements, architectural design, **system and database modeling**; built **APIs**, mobile, and web applications with a cross-functional team
- Designed containerized **microservices** on EKS and ECS with autoscaling, load balancing, and seamless inter-service communication
- Reduced peak latency by **90% through database optimizations**: query tuning, caching layers, and indexing strategy redesign
- Built a **high-throughput background processing system** handling 1M+ daily jobs (payments, messaging, notifications) with provider failover
- Automated **infrastructure provisioning** with Terraform, reducing setup time by 90% while building CI/CD pipelines
- Designed **disaster recovery**: parallel GCP environment mirroring AWS using GKE, Cloud SQL, with dual-environment CI/CD for rapid failover

PROJECTS

Virk Cloud -- virk.cloud Infrastructure Platform

Architected a production-grade Kubernetes PaaS enabling startups to deploy clusters instantly without operational overhead. Built-in observability stack for centralized monitoring, real-time log streaming, and alerts. Automated all deployments using FluxCD and Terraform for reproducible, version-controlled operations on bare-metal VPS with RKE2.

RKE2, FluxCD, Terraform, Prometheus, Grafana

AI Agents for Infrastructure Operations Current

Building specialized AI agents for QA automation and sysadmin operations on production Linux servers. Developing agent-driven SaaS products leveraging RAG, tool-use, and multi-agent orchestration patterns for infrastructure automation.

Python, LLM APIs, RAG, Multi-Agent Orchestration

Education Research and Innovation Foundry -- erifoundry.org Research

Leading design of AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics. Coordinating cross-disciplinary research for scalable ML solutions in education.

LLMs, ML, Research Frameworks

EDUCATION

BSc Computer Engineering -- Ashesi University

Sep 2016 - May 2020

Berekuso, Ghana

CERTIFICATIONS

Oracle -- Certified DevOps Professional

Oracle -- Certified Cloud Solutions Architect Associate

ByteByteAI -- AI Engineering Specialization (Agents, RAG, LLM Applications)

SKILLS

Cloud & Infrastructure: AWS, Azure, GCP, OCI, VMware, On-Premise/Bare Metal Data Centers (Tier III), Linux Administration

Container Orchestration: Kubernetes (EKS, GKE, Tanzu/TKG, RKE2), Docker, Helm, Istio Service Mesh

CI/CD, GitOps & IaC: Terraform, Ansible, Bash, Python, GitHub Actions, ArgoCD, FluxCD, Azure DevOps

Observability: Prometheus, Grafana, ELK Stack, Sentry

Networking & Security: IPSec/OpenVPN, TLS/SSL, Firewalls (WAF), OAuth2, Data Sovereignty Compliance

Architecture: Microservices Design, API Design, Database Modeling, Disaster Recovery, Multi-Cloud Architecture

AI/Agents: LLM Integration, RAG Systems, AI Agent Development for Ops Automation